L. Ruth Rivkin

CURRICULUM VITAE

SUMMARY OF CV

Research and Teaching Statement

I am a broadly trained bioinformatician studying the ecology and evolution of populations experiencing climate change. I use cutting edge bioinformatic tools and AI-driven data analysis to inform conservation and management decisions for vulnerable Arctic species. I have published my award-winning research in leading journals and have extensive experience teaching undergraduate courses.

Summary of Excellence in Research and Teaching

- 21 manuscripts published in peer-reviewed journals, 2 in review
- $350,000 in external fellowship and award funding
- 7 invited talks and seminars, 3 best talk awards
- Course instructor at University of Toronto Mississauga (4.5/5 course rating)
- 2 undergraduate research projects supervised

ACADEMIC HISTORY

Research Appointments

Weston Family Fellow in Northern Research 2026-2028
Adaptation and resilience of polar bears to climate change
University of Manitoba, Polar Bears International

NSERC Postdoctoral Research Fellow 2022-2026
Genomic basis of adaptation in polar bears threatened by climate change
University of Manitoba, Polar Bears International, and San Diego Zoo Wildlife Alliance

Data Scientist (Contractual) 2023
Genetically-informed species distribution modeling
Environment and Climate Change Canada

Postdoctoral Researcher 2021-2022
Conservation genomics of polar bears under climate change
Polar Bears International and San Diego Zoo Wildlife Alliance

Career Interruptions

Parental leave (1 year) 2025-2026
Parental leave (1 year) 2022-2023

Education

Ph.D. Ecology and Evolutionary Biology 2015-2021
University of Toronto

Thesis: The effects of urbanization on the ecology and evolution of species interactions
Supervisor: Marc Johnson

Hon. B.Sc. Wildlife Biology-Statistics 2010-2015
University Guelph

SCHOLARLY AND PROFESSIONAL WORK

Refereed Publications *(Total = 21; h = 13; Citations = 1232)*

21. **Rivkin LR**, K King, J Aars, K Laidre, A Langwieder, D McGeachy, L Newediuk, S Petersen, N Pilfold, AP Wilder, G York, ES Richardson, C Garroway. 2026. Climate-linked evolution and genetics in a warming Arctic. In press at *Ecological Monographs*. Submission ID ECM25-0070.R1. https://doi.org/10.32942/X27K92
20. **Rivkin LR**, C Garroway, MTJ Johnson. 2026. A multi-city assessment of genomic evolution in the native wildflower *Impatiens capensis*. Accepted at *Evolutionary Applications*. Submission ID 4812859. https://doi.org/10.22541/au.173884755.59553944/v1
19. de Greef E, C Müller, AA Snead, **LR Rivkin**, SH Ferguson, CA Watt, M Marcoux, SD. Petersen, CJ Garroway. 2026. Identifying areas of potential risk based on future genetic adaptability in three Arctic whale species. *American Naturalist*. 207: 347-465. https://doi.org/10.1086/738889
18. **Rivkin LR**, ES Richardson, JM Miller, TC Atwood, S Baryluk, EW Born, C Davis, M Dyck, E de Greef, KL Laidre, NJ Lunn, S McCarthy, ME Obbard, MA. Owen, NW Pilfold, A Roberto-Charron, Ø Wiig, AP Wilder, CJ Garroway. 2024. Assessing the risk of climate maladaptation for Canadian polar bears. *Ecology Letters*. 27:e14486. https://doi.org/10.1111/ele.14486
17. Miller JM, RM Malenfant, **LR Rivkin***, TC Atwood, S Baryluk, EW Born, R Dietz, KL Laidre, J Pongracz, ES Richardson, Ø Wiig, CS Davis. 2024. Development of an 8K SNP chip to assess adaptive diversity and hybridization in polar bears. *Conservation Genetics Resources*. 16:237-249. https://doi.org/10.1007/s12686-024-01359-1

**Corresponding author*

16. Urban, MC, M Alberti, L De Meester, Y Zhou, BC Verrelli, M Szulkin, C Schmidt, P Roberts, **LR Rivkin**, EP Palkovacs, J Munshi-South, A Melesis, NC Harris, KM Gotanda, CJ Garroway, SE Diamond, S Des Roches, A Charmantier, KI Brans. 2024. Interactions between climate change and urbanization will shape the future of biodiversity. *Nature Climate Change*. 14:436–447. https://doi.org/10.1038/s41558-024-01996-2
15. **Rivkin LR**, AC Andrade. 2023. Increased herbivory but not cyanogenesis is associated with urbanization in a tropical wildflower. *Austral Ecology*. 48:388-398 https://doi.org/10.1111/aec.13274
14. **Rivkin LR**, MTJ Johnson. 2022. The impact of urbanization on mating system and population genetic variation in the native wildflower, *Impatiens capensis*. *Journal of Urban Ecology*. 8: juac009. https://doi.org/10.1093/jue/juac009
13. Verrelli, BC, A Alberti, S Des Roches, NC Harris, AP Hendry, MTJ Johnson, AM Savage, A Charmantier, KM Gotanda, L Govaert, LS Miles, **LR Rivkin**, KM Winchell, KI Brans, Cn Correa, SE Diamond, B Fitzhugh, NB Grimm, S Hughes, JM Marzluff, J Munshi-South, C Rojas, JS Santangelo, CJ Schell, JA Schweitzer, M Szulkin, MC Urban, Y Zhou, C Ziter. 2022. A global

horizon scan for urban evolutionary ecology. *Trends in Ecology and Evolution.* 37:1006-1019. https://doi.org/10.1016/j.tree.2022.07.012

12. KM Winchell, KJ Aviles-Rodriguez, EJ Carlen, LS Miles, A Charmantier, LF De León, KM Gotanda, **LR Rivkin**, M Szulkin, and BC Verrelli. 2022. Moving past the challenges and misconceptions in urban adaptation research. *Ecology and Evolution*. 12:ee9552. http://doi.org/10.1002/ece3.9552
11. **Rivkin LR**, RA Johnson, JA Chaves, MTJ Johnson. 2021. Urbanization alters ecological and evolutionary interactions between Darwin's finches and *Tribulus cistoides* on the Galápagos Islands. *Ecology and Evolution*. 11:15754–15765. http://doi.org/10.1002/ece3.8236
10. **Rivkin, LR**, V Nhan[+], AE Weis, MTJ Johnson. 2020. Variation in pollinator-mediated plant reproduction across an urbanization gradient. *Oecologia*. 192:1073–1083. https://doi.org/10.1007/s00442-020-04621-z

[+] *Undergraduate mentee*

9. Des Roches, S, K Brans, M Lambert, **LR Rivkin**, A Savage, C Schell, C Correa, L De Meester, S Diamond, N Grimm, N Harris, A Hendry, L Goevart, MTJ Johnson, J Munshi-South, E Palkovacs, M Szulkin, M Urban, B Verrelli, M Alberti. 2020 Socio-eco-evolutionary dynamics in cities. *Evolutionary Applications*. 14:248– 267. https://doi.org/10.1111/eva.13065
8. Santangelo JS, **LR Rivkin**, C Advenard, KA Thompson. 2020 Multivariate phenotypic divergence along an urbanization gradient. *Biology Letters*. 16:20200511. https://doi.org/10.1098/rsbl.2020.0511
7. Santangelo, JS, LS Miles, ST Breitbart, D Murray-Stoker, **LR Rivkin**, MTJ Johnson, RW Ness. 2020. Urban environments as a framework to study parallel evolution. *In* Szulkin, M, Munshi-South, J, Charmantier, A, eds. Urban Evolutionary Biology. Oxford University Press.
6. **Rivkin, LR**, JS Santangelo, M Alberti, MF Aronson, CW de Keyzer, SE Diamond, M Fortin, LJ Frazee, AJ Gorton, AP Hendry, Y Liu, JB Losos, JS MacIvor, RA Martin, M McDonnell, LS Miles, J Munshi-South, R Ness, AE Newman, MR Stothart, P Theodorou, KA Thompson, BV Verrelli, A Whitehead, KM Winchell, and MTJ Johnson. 2019. A roadmap for urban evolutionary ecology. *Evolutionary Applications.* 12:384–398. https://doi.org/10.1111/eva.12734
5. Miles, LM, **LR Rivkin***, MTJ Johnson, J Munshi-South, BC Verrelli. 2019. Gene flow and genetic drift in urban environments. *Molecular Ecology*. 28:4138-4151. https://doi.org/10.1111/mec.15221

**Co-first author*

4. **Rivkin, LR**, SCH Barrett, MTJ Johnson. 2018. The effects of plant sexual system and latitude on resistance to herbivores. *American Journal of Botany*. 105:977-985. https://doi.org/10.1002/ajb2.1098
3. Santangelo, JS, **LR Rivkin**, MTJ Johnson. 2018. The evolution of city life. *Proceedings of the Royal Society B*. 20181529. https://doi.org/10.1098/rspb.2018.1529
2. **Rivkin, LR**, AL Case, CM Caruso. 2016. Why is gynodioecy a rare but widely distributed sexual system? Lessons from the Lamiaceae. *New Phytologist*. 211:688-696. https://doi.org/10.1111/nph.13926
1. **Rivkin, LR**, AL Case, CM Caruso. 2015. Frequency-dependent fitness in gynodioecious *Lobelia siphilitica*. *Evolution*. 69:1232-1243. https://doi.org/10.1111/evo.12654

Manuscripts in Review *(Total:2)*

1. Newediuk L, ES Richardson, B Biddlecombe, H Kassim, L Kathan, N Lunn, **LR Rivkin**, O Salama, C Schmidt, M Jones, CJ Garroway. Climate change, age acceleration, and the erosion of fitness in polar bears. In revision at *Nature*. Submission ID 2024-02-02361C. https://doi.org/10.1101/2024.01.05.574416
2. Kess T, M Van Wyngaarden, S Beck, D Davenport, C Nugent, B Marquis, S Lehnert, H Arno, **LR Rivkin**, AM Fijarczyck, P Lenz, N Isabel, I Bradbury. Are we overcorrecting for population structure? Rethinking how to detect and predict the genome-environment relationship across species. In review at *Molecular Ecology Resources*. Submission ID 6760025.

Refereed Special Issues as Guest Editor

Santangelo, JS, **LR Rivkin**, MTJ Johnson. 2018. The evolution of city life. Special Issue in *Proceedings of the Royal Society B*.

Honours, Awards, and Scholarships (*Total funding: $350,000; Academic (A); Leadership (L))*

Weston Family Postdoctoral Fellowship in Northern Studies ($110,000; A)	2026-2028
McCall MacBain 10 Minute Postdoctoral Research Talks Finalist (A)	2024
NSERC Postdoctoral Research Fellowship ($90,000; A)	2022-2026
UTM Graduate Student Research Award ($2,500; A)	2021
EEB Department Harold Harvey Award Honorable Mention ($250; L)	2021
UTM Roberta Bondar Graduate Student Excellence Award ($1,126.43; L/A)	2020
CSEE Excellence in Doctoral Research Award ($500; L/A)	2020
EEB Department Peter Abrams Award Honorable Mention ($250; A)	2020
QEII Graduate Scholarship in Science and Technology ($15,000; A)	2019-2020
EEB Department Ramsay-Wright Award Honorable Mention ($250; A)	2019
UTM Graduate Training Award ($500; A)	2018
NSERC Canada Graduate Scholarship Doctoral ($105,000; A)	2016-2019
Sigma Xi Grant-in-Aid of Research ($900; A)	2016
Best Graduate Talk, CSEE ($500; A)	2016
UTM Conference Travel Grant ($400/year; A)	2016-2019
UTM Biology Travel Award ($1000/year; A)	2016-2019
NSERC Canada Graduate Scholarships – Master's Program ($17,500; A)	2015-2016
Botanical Society of America Young Botanist Award (A)	2015
Best Undergraduate Talk, OE3C ($150; A)	2014

Papers Presented at Meetings, Symposia, and Institutions *(Total: 23)*

Invited Talks and Seminars

1. **Rivkin LR**. February 2025. Polar bear evolution and genetics in a warming Arctic. Polar Bear Technical Committee. Invited Speaker. Nain, Labrador
2. **Rivkin LR**. 2024. Can polar bears adapt to climate change? Insights from genetics. San Diego Zoo Wildlife Alliance Postdoctoral Seminar Series. Virtual.

3. **Rivkin LR**. 2023. Conservation genetics of polar bears in the Canadian Arctic. Government of Canada Genomic Resilience and Adaptation to Climate Change Project Seminar Series. Virtual.
4. **Rivkin LR**. 2021. The effects of urbanization on the ecology and evolution of species interactions University of Toronto Mississauga Biology Department Seminar Series. Virtual
5. **Rivkin LR**. 2020. The effects of urbanization on the ecology and evolution of species interactions. Doctoral Excellence Award Talk. Canadian Society for Ecology and Evolution. Virtual.
6. **Rivkin LR**. 2020. A roadmap for urban evolutionary ecology. Caribaea Initiative Research & Conservation Conference. Bridgetown, Barbados. Plenary. *Cancelled due to COVID-19*
7. **Rivkin LR**, V Nahn, AE Weis, MTJ Johnson. 2019. The strength of plant-pollinator interactions varies along an urbanization gradient. Urban Eco-Evolutionary Dynamics Workshop. Seattle, WA.

Contributed Conference Presentations (Oral presentation (T), Poster presentation (P))

1. **Rivkin LR**, ES Richardson, CJ Garroway. 2024. Assessing the risk of climate maladaptation for Canadian polar bears. International Association for Bear Research and Management (IBA) 28th Conference. Edmonton, AB. (P)
2. **Rivkin LR**. 2024. Assessing the risk of climate maladaptation for Canadian polar bears. 3rd Joint Congress on Evolutionary Biology. Montreal, QC. (T)
3. **Rivkin LR***. 2024. Can polar bears adapt to climate change? Insights from genetics. McCall MacBain 10 Minute Postdoctoral Research Talks. Hamilton, ON. (T) Recorded.

**Finalist*

4. **Rivkin LR**. 2023. Conservation genetics of polar bears in the Canadian Arctic. International Polar Bear Genetics Meeting. Winnipeg, MB. (T)
5. **Rivkin LR**. Richardson, E, Garroway, C. 2023. Genetic vulnerability of polar bears to sea ice loss. Canadian Society for Ecology and Evolution. Winnipeg, MB. (T)
6. **Rivkin LR**, RA Johnson, JA Chaves, MTJ Johnson. 2021. Urbanization alters ecological and evolutionary interactions between Darwin's finches and *Tribulus cistoides* on the Galápagos Islands. Society for the Study of Evolution (Evolution). Virtual. (T)
7. **Rivkin LR**, V Nahn, AE Weis, MTJ Johnson. 2019. The strength of plant-pollinator interactions varies along an urbanization gradient. Society for the Study of Evolution (Evolution). Providence, RI. (T)
8. Miles LS, **Rivkin LR**, Johnson MTJ, J Munshi-South, BC Verrelli. 2020. Gene flow and genetic drift in urban environments. Evolution. Providence, RI. (T)
9. **Rivkin LR**, V Nhan, MTJ Johnson. 2018. The influence of urbanization on plant fitness and pollinator communities. II Joint Congress on Evolutionary Biology. Montpellier, FR. (P)
10. **Rivkin LR**, MTJ Johnson. 2017. The role of urbanization on the evolution of selfing and population genetic structure in *Impatiens capensis*. Evolution. Portland, OR. (P)
11. **Rivkin LR***, SCH Barrett, MTJ Johnson. 2016. The role of sexual system and latitude on insect herbivory in *Sagittaria latifolia* (Alismataceae). Canadian Society for Ecology and Evolution. St. John's, NL. (T)

**Best Graduate Talk*

12. **Rivkin LR**, SCH Barrett, MTJ Johnson. 2016. The role of sex-biased herbivory on the evolution of dioecy in *Sagittaria latifolia* (Alismataceae). Ontario Ecology, Ethology and Evolution Colloquium. Toronto (OE3C), ON. (T)
13. **Rivkin LR**, AL Case, CM Caruso. 2015. Why is gynodioecy a rare but widespread sexual system? Lessons from the Lamiaceae. OE3C. Toronto, ON. (T)
14. **Rivkin LR**, AL Case, CM Caruso. 2014. Mechanisms of frequency dependent fitness in gynodioecious *Lobelia siphilitica*. Evolution. Raleigh, NC. (T)
15. **Rivkin LR***, AL Case, CM Caruso. 2014. Mechanisms of frequency dependent fitness in gynodioecious *Lobelia siphilitica*. OE3C. (T)
    **Best Undergraduate Talk*

TEACHING AND SUPERVISORY EXPERIENCE

Undergraduate Course Instructor

Evolutionary Biology (BIO342; 150 students; sole instructor)	2021

Teaching Assistantships and Course Development

Teaching Assistantships

Ecology (BIO205)	2020
Evolution (BIO152)	2019
Biology behind the news (BIO201)	2018
Ecology (BIO205)	2017
Evolutionary Biology (BIO342)	2016-2020
Biometrics II (BIO361)	2015-2018

Course Development

Biometrics II (BIO361)	2019

Supervisory Experience

Vanessa Nhan, Honors Thesis (BIO481)	2018
Zahra Sina, Research Project (BIO299)	2016

ACADEMIC SERVICE

Academic and Community Service

Symposium organizer, Conservation Genetics Across Multiple Species Symposium 3rd Joint Congress of Evolutionary Biology	2024
Organizer; International Polar Bear Genetics Meeting	2023
Steering Committee; NSF-RCN: Urban Eco-Evo Dynamics	2022-2023
Advisory Council, NSF LTER Urban Evolution Workshop	2021-2022
Lead organizer, NSF-RCN: Urban Eco-Evo Dynamics Workshop Series	2021-2022
Panel Judge, Global Undergraduate Awards	2020
Volunteer, St. John's Ambulance Therapy Dog Program	2020
Graduate student representative, Canadian Society for Ecology and Evolution	2018-2020

Co-organizer, The evolution of city life. Special Issue in *Proceedings of the*

Royal Society B.	2017-2018
Executive, Synthesis in the City: Urban Evolutionary Ecology of Plants Workshop Organizing Committee	2016-2017
Executive, OE3C	2015-2016

Departmental and University Service

Graduate representative, UTM Biology Executive Committee	2020
Graduate Representative, UTM Health and Counselling Committee	2019
Founder, Path to Resilience: UTM Graduate Peer Support Group	2018-2020
Organizer, Path to Resilience: UTM Grad Wellness Workshop Series	2018-2020
Treasurer, UTM Association of Graduate Students	2018-2020
President, UTM Biology Graduate Student Society	2016-2017
Graduate representative, UTM Biology Executive Committee	2016-2017
Graduate representative, UTM Biology faculty meetings	2016-2017
Graduate representative, IT Student Advisory Committee	2015-2016

External reviews

Conservation Genetics	New Phytologist	Frontiers in Ecology and Evolution
Ecology and Evolution	Urban Ecosystems	Industrial Crops and Products
Heredity	American Journal of Botany	Journal of Evolutionary Biology
Scientific Reports	Evolutionary Applications	Plant Ecology
Journal of Ecology	PeerJ	Botany
Bioscience	New Phytologist	Biological J of the Linnean Society
Evolution Letters	Molecular Ecology	Genetica

RELEVANT EXPERIENCE

Media and Public Appearances

Radio and Television Interviews

Grolar bears and climate change. CBC Radio syndicated interviews (13 afternoon talk show interviews).

Arctic Hybrids. 2024. Global News 640 Toronto This Weekend with Ben Mulroney.

How common at grolar bears? 2024. CBC Radio. https://www.cbc.ca/player/play/video/9.6432675

Polar Bear Science Steps Forward. 2023. Tundra Connections. Guest Expert. https://www.youtube.com/watch?v=OzHztaAAPwA

City Animals. 2022. Frick, I love Nature. Season 1, Episode 2. CBC Gem. Guest Expert. https://gem.cbc.ca/media/frick-i-love-nature/s01e02

Why house sparrows can eat starch: the evolution of city wildlife. 2018. The Environment Report. Michigan NPR Radio. https://www.michiganradio.org/post/why-house-sparrows-can-eat-starch-evolution-city-wildlife

Print Media

Study paints grim picture of polar bear survival in the Arctic. 2024. CTV Winnipeg. https://winnipeg.ctvnews.ca/mobile/study-paints-grim-picture-of-polar-bear-survival-in-the-arctic-1.6992299?cache=?autoPlay=true/7.319866/7.319866/7.319866/7.393958

Certains ours polaires de l'Arctique s'adaptent mieux que d'autres, selon une étude [Some polar bears in the Arctic are adapting better than others, study finds]. 2024. Radio-Canada. https://ici.radio-canada.ca/nouvelle/2096520/population-decroissante-arctique-glace-recherche

Some polar bears may be able to survive climate change, U of M researcher reports in co-authored study. 2024. Winnipeg Free Press. https://www.winnipegfreepress.com/breakingnews/2024/08/08/some-polar-bears-may-be-able-to-survive-climate-change-u-of-m-researcher-reports-in-co-authored-study

Polar bears in Canada's high Arctic are most vulnerable to climate change, research shows. 2024. Winnipeg Sun. https://winnipegsun.com/news/provincial/polar-bears-in-canadas-high-arctic-are-most-vulnerable-to-climate-change-research-shows

Pizzlies, grolars, and narlugas: Why we may soon see more Arctic hybrids. 2024. National Geographic https://www.nationalgeographic.com/animals/article/arctic-hybrid-polar-grizzly-bear-genetics-climate?loggedin=true&rnd=1723027142398

'Grolar' hybrid of grizzlies and polar bears remains rare in wild, study finds. 2024. The Guardian. https://www.theguardian.com/environment/article/2024/jun/20/grolar-hybrid-grizzly-polar-bears-study

Grolar bear hybrids in N.W.T. all traced back to same 'strange' female polar bear. 2024. CBC North. https://www.cbc.ca/news/canada/north/polar-bear-grizzly-study-1.7239850

Grizzly-polar bear hybrids extremely rare but predicted to rise with global warming, say scientists. 2024. BBC Wildlife Magazine. https://www.discoverwildlife.com/animal-facts/pizzly-grolar-bears

Unchecked climate change could render Hudson Bay polar bears extinct by 2030s, report warns. 2024. Winnipeg Free Press. https://www.winnipegfreepress.com/breakingnews/2024/06/13/unchecked-climate-change-could-render-hudson-bay-polar-bears-extinct-by-2030s-report-warns

'Grolar' hybrid bears confirmed in Canadian study. 2024. Times Colonist. https://www.timescolonist.com/animal-stories/grolar-hybrid-bears-confirmed-in-canadian-study-9080567

Hybridization Extremely Rare Between Grizzly and Polar Bears, Study Finds. 2024. Polar Bears International. https://polarbearsinternational.org/news-media/articles/hybridization-extremely-rare-between-grizzly-and-polar-bears-study-finds

Grolar Bear Hybrids Are "Extremely Rare", With Just 8 Confirmed Individuals. IFL Science.https://www.iflscience.com/grolar-bear-hybrids-are-extremely-rare-with-just-8-confirmed-individuals-74652

UM researcher helps to track Grolar or Pizzly bears. 2024. UM News Today. https://news.umanitoba.ca/um-researcher-helps-to-track-grolar-or-pizzly-bears/

Decoding Polar Bears by Ruth Rivkin. 2023. Polar Bears International Annual Newsmagazine. https://polarbearsinternational.aflip.in/2023newsmagazine#page/27

Can polar bears adapt to climate change? Insights from genetics. 2024. Appetizers for the Mind. Department of Psychology, Neuroscience, and Behaviour, McMaster University. Hamilton, ON.

Can polar bears adapt to climate change? Insights from genetics. 2024. Royal City Men's Club. Guelph, ON.

Ruth Rivkin wins Excellence in PhD Research Award. 2020. U of T Celebrates. https://www.utoronto.ca/celebrates/ruth-rivkin-wins-excellence-phd-research-award

Research Experience

Technician, EvoEco Lab 2015
University of Toronto Mississauga (Dr. Marc Johnson)

Technician, Plant Evolutionary Ecology Research 2013-2015
University of Guelph (Dr. Christina Caruso)

Technician, Marine Benthic Lab Research 2012, 2010
Memorial University (Dr. Paul Snelgrove)

Technician, Laboratoire Environmental de Géochimie Organique Research 2011
Concordia University (Dr. Yves Gelinas)